

AETHERIA — Tech Stack

Step 2 · choices + rationale + version pin policy

1. Frontend

Concern	Choice	Why
Framework	Next.js 15 (App Router)	SSR for SEO menus, RSC, route-based code-split
Language	TypeScript 5.6	end-to-end type safety with backend
UI lib	React 19	Server Components, transitions
Styling	Tailwind CSS 4 + shadcn/ui	utility-first, theme tokens
Game canvas	PixiJS 8 (WebGL/WebGPU)	60 fps 2.5D, sprite batching, filters
Audio	Howler.js 2	spatial audio, sprite-sheet sounds
State	Zustand 5 + TanStack Query	minimal boilerplate, server cache
Forms / validation	react-hook-form + Zod	shared schema with backend
i18n	next-intl	route-based locale, ICU
PWA / Offline	next-pwa + idb	install-to-home, IndexedDB cache
Animations	framer-motion	menu transitions

2. Backend

Concern	Choice	Why
Runtime	Node.js 22 LTS	stable LTS
HTTP framework	Fastify 5	fast, schema-first, plugin model
API protocol	tRPC 11	end-to-end types, no codegen
Realtime	Socket.IO 4 + Redis adapter	rooms, broadcasting, horizontal scale
Auth	NextAuth.js v5 (Auth.js)	OAuth + email magic + JWT
ORM	Prisma 5	type-safe, migrations, introspection
Validation	Zod 3	shared with client
Background jobs	BullMQ 5	Redis-backed cron + queues
Logging	pino	fast structured logs
Tracing	OpenTelemetry SDK	vendor-neutral
Rate limit	@fastify/rate-limit	per IP + per user
Email	Resend	transactional + verification

3. Data

Layer	Choice	Why
RDBMS	PostgreSQL 16	JSONB for flexible level/payload data
Cache	Redis 7	sessions, leaderboards (ZSET), pub/sub
Object store	Cloudflare R2 (S3-compatible)	cheap, no egress fee
Search	Postgres FTS (later: Meilisearch)	start simple
Analytics	ClickHouse (optional later)	event funnels, retention
Schema mgmt	Prisma Migrate	single source of truth
Backup	Neon PITR (7 days) + nightly logical to R2	RPO ≤ 5 min

4. DevOps

Concern	Choice
Package manager	pnpm 9 (workspace)
Monorepo	Turborepo 2
Lint	ESLint 9 (flat) + Prettier 3
Pre-commit	lefthook
CI	GitHub Actions
CD	Vercel (web) + Fly.io (api/realtime/worker)
Container	Docker (multi-stage)
Secrets	Doppler (dev) + provider env (prod)
Testing	Vitest (unit) + Playwright (e2e) + Pact (contract)
Coverage	vitest --coverage (target C0=100%, C1>90% per spec)
Error monitoring	Sentry
Feature flags	OpenFeature + LaunchDarkly Flagsmith fallback

5. Why NOT (rejected alternatives)

- **Unity / Unreal** → heavyweight; web build is awkward; we want PWA-first.
- **Phaser** → fine for 2D, but PixiJS gives better fine-grained control + smaller bundle.
- **MongoDB** → relational integrity matters (inventory, ranking); JSONB in Postgres covers flex needs.
- **Express** → slower; Fastify is the modern default.
- **GraphQL** → overhead unjustified; tRPC fits a single TS team better.
- **Kafka** → too heavy for current scale; BullMQ + Redis pubsub is enough until 100 k DAU.

6. Version pin policy

- All major versions pinned in `package.json` (no `^` for majors).
- Renovate bot opens PRs weekly; CI must pass before merge.
- Node + pnpm versions pinned in `.nvmrc`, `package.json#packageManager`, and Dockerfile.

7. Browser support matrix

Browser	Min version	Notes
Chrome	120	primary
Edge	120	
Firefox	120	
Safari	17	iOS audio quirks tested
Mobile Chrome / Safari	latest 2	Touch controls